Chapter 4. Double Integrals

Contents

4.1 Product Measures **2**

4.1.1 Setting up the problem . . . 2

4.1.2 The smallest σ-algebra . . . 2

4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$ . . . 2

4.1.4 The monotone class argument . . . 3

4.1.5 The product of Lebesgue measures; Borel sets . . . 5

4.1.6 Lebesgue measure on $\mathbb{R}^2$ . . . 6

4.2 Fubini's Theorem **7**

4.2.1 Borel functions . . . 9

4.3 Convolution of Integrable Functions **11**

4.3.1 Definition . . . 11

4.3.2 Convolution of L^p and L^q functions . . . 13

4.3.3 Averaging kernels . . . 14

Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). Interchanging the order of integration needs a measure on the plane. This chapter defines the **product** of two measures, the double integral, the conditions under which it equals the iterated integrals (Tonelli, Fubini), and the **convolution**.

4.1 Product Measures

Motivation. Iterated integration needs a measure on $X \times Y$ built from those on X and Y, defined on sets far more complicated than rectangles; hence the monotone class argument.

4.1.1 Setting up the problem

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be measure spaces. A σ-algebra on $X \times Y$ should contain the **measurable rectangles**

$$\mathcal{R} = \{S \times T \subset X \times Y : \ S \in \mathcal{S}, \ T \in \mathcal{T}\}.$$

For $E \subset X \times Y$, $x \in X$, $y \in Y$ define the **sections**

$$E_x = \{y \in Y : (x, y) \in E\}, \qquad E^y = \{x \in X : (x, y) \in E\}.$$

If $E_x \in \mathcal{T}$ and $E^y \in \mathcal{S}$ for all x, y, let

$$\varphi : x \mapsto \nu(E_x) = \int_Y \chi_E(x, y)\, d\nu(y), \qquad \psi : y \mapsto \mu(E^y) = \int_X \chi_E(x, y)\, d\mu(x). \tag{1}$$

If φ, ψ are measurable and

$$\int_X \varphi \, d\mu = \int_Y \psi \, d\nu, \tag{2}$$

this common value should be the measure of E. We show that every E in the smallest σ-algebra $\mathcal{S} \times \mathcal{T}$ containing $\mathcal{R}$ has these properties.

4.1.2 The smallest σ-algebra

Let $\mathcal{U}$ be a collection of subsets of Z and $\Omega \neq \emptyset$ the family of σ-algebras containing $\mathcal{U}$ (the power set is one). Put

$$\mathcal{W} = \bigcap \{\mathcal{M} : \mathcal{M} \in \Omega\}.$$

$\emptyset \in \mathcal{W}$; if $A \in \mathcal{W}$ then $A^c \in \mathcal{M}$ for every $\mathcal{M} \in \Omega$, so $A^c \in \mathcal{W}$; likewise for countable unions. So $\mathcal{W}$ is a σ-algebra contained in every σ-algebra containing $\mathcal{U}$: the **smallest σ-algebra containing $\mathcal{U}$**.

4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$

Let

$$\mathcal{A} = \{E \in \mathcal{S} \times \mathcal{T} : \ E_x \in \mathcal{T} \ (x \in X)\}.$$

$\mathcal{R} \subset \mathcal{A}$, so it suffices that $\mathcal{A}$ is a σ-algebra: $\emptyset \in \mathcal{A}$, $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$. Treating E^y likewise,

$$E \in \mathcal{S} \times \mathcal{T} \implies E_x \in \mathcal{T}, \ E^y \in \mathcal{S} \qquad (x \in X, \ y \in Y). \tag{3}$$

Exercise 4.1.1

In the argument above, show that $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$.

Solution [Supplement]. $y \in (E^c)_x \iff (x,y) \in E^c \iff (x,y) \notin E \iff y \notin E_x \iff y \in (E_x)^c$. $y \in (\bigcup_n E_n)_x \iff (x,y) \in E_n$ for some $n \iff y \in (E_n)_x$ for some $n \iff y \in \bigcup_n (E_n)_x$. □

4.1.4 The monotone class argument

Assume from now on that

$$X = \bigsqcup_{n=1}^{\infty} X_n, \qquad Y = \bigsqcup_{m=1}^{\infty} Y_m, \qquad \mu(X_n),\ \nu(Y_m) < \infty.$$

To show (2) for all $E \in \mathcal{S} \times \mathcal{T}$, let

$$\mathcal{B} = \{E \in \mathcal{S} \times \mathcal{T} :\ \text{the functions } \varphi, \psi \text{ of (1) are measurable and (2) holds}\},$$
$$\mathcal{C} = \{E \in \mathcal{S} \times \mathcal{T} :\ E \cap (X_n \times Y_m) \in \mathcal{B}\ \ (m, n = 1, 2, \dots)\}.$$

Lemma 4.1.1

For the collections $\mathcal{B}$ and $\mathcal{C}$:

(a) $\mathcal{B}$ is closed under unions of increasing sequences.

(b) $\mathcal{B}$ is closed under countable disjoint unions.

(c) $\mathcal{C}$ is closed under unions of increasing sequences and intersections of decreasing sequences.

(d) If $E_1, \dots, E_N \in \mathcal{R}$ are pairwise disjoint, then $\bigcup_{n=1}^{N} E_n \in \mathcal{C}$.

Proof. (a) Let $E_i \in \mathcal{B}$ increase to E, with functions φ_i, ψ_i and φ, ψ. As $(E_i)_x \nearrow E_x$, Fact B.24(b) gives $\varphi_i \nearrow \varphi$, so φ is measurable; likewise ψ. By monotone convergence

$$\int_X \varphi \, d\mu = \lim_{i\to\infty} \int_X \varphi_i \, d\mu = \lim_{i\to\infty} \int_Y \psi_i \, d\nu = \int_Y \psi \, d\nu.$$

(b) For disjoint $E, F \in \mathcal{B}$ the functions in (1) add, so $\mathcal{B}$ is closed under finite disjoint unions; a countable disjoint union is an increasing union of finite ones, so use (a).

(c) If $E_i \in \mathcal{C}$ increase, then for each m, n

$$\langle E_i \cap (X_n \times Y_m) : i = 1, 2, \dots \rangle \tag{4}$$

increases in $\mathcal{B}$, so $\bigcup_i E_i \in \mathcal{C}$ by (a). For decreasing E_i, argue as in (a) with Fact B.24(c) (as $\nu(Y_m), \mu(X_n) < \infty$) and dominated convergence (bound $\nu(Y_m)\chi_{X_n}$).

(d) $\mathcal{R} \subset \mathcal{C}$: for $E = S \times T$, $\varphi = \nu(T \cap Y_m)\chi_{S \cap X_n}$ and $\psi = \mu(S \cap X_n)\chi_{T \cap Y_m}$ have equal integrals $\mu(S \cap X_n)\nu(T \cap Y_m)$. Then use (b). ■

Let $\mathcal{D}$ be the smallest collection containing $\mathcal{R}$ with (c) and (d) of Lemma 4.1.1. Then

$$\mathcal{D} \subset \mathcal{C} \subset \mathcal{S} \times \mathcal{T}, \tag{5}$$

and we show $\mathcal{D}$ is a σ-algebra. With $\mathcal{E}$ the finite disjoint unions of measurable rectangles, $\mathcal{D}$ is the smallest *monotone class* (closed under increasing unions and decreasing intersections) containing $\mathcal{E}$.

Exercise 4.1.2

Show that for any collection of sets there is a smallest collection containing it that is closed under unions of increasing sequences and intersections of decreasing sequences.

Solution [Supplement]**.** With $\Omega \neq \emptyset$ the monotone classes containing $\mathcal{U}$ (the power set is one), $\mathcal{W} = \bigcap \Omega$ contains $\mathcal{U}$, and a monotone sequence in $\mathcal{W}$ is one in every $\mathcal{M} \in \Omega$, so its limit is in $\mathcal{W}$. So $\mathcal{W}$ is the smallest monotone class containing $\mathcal{U}$. □

$\mathcal{E}$ is closed under differences and unions:

$$E, F \in \mathcal{E} \implies E \setminus F \in \mathcal{E},\ F \setminus E \in \mathcal{E},\ E \cup F \in \mathcal{E}. \tag{6}$$

Exercise 4.1.3

Prove (6).

Solution [Supplement]**.** (i) $(S \times T) \cap (S' \times T') = (S \cap S') \times (T \cap T')$, so for $E = \bigsqcup_i R_i$, $F = \bigsqcup_j R'_j$ in $\mathcal{E}$, $E \cap F = \bigsqcup_{i,j}(R_i \cap R'_j) \in \mathcal{E}$.

(ii) $(S \times T)^c = (S^c \times Y) \sqcup (S \times T^c)$, so $(\bigsqcup_i R_i)^c = \bigcap_i R_i^c \in \mathcal{E}$ by (i).

(iii) So $E \setminus F = E \cap F^c$, $F \setminus E \in \mathcal{E}$ and $E \cup F = (E \setminus F) \sqcup F \in \mathcal{E}$. □

For an arbitrary $E \subset X \times Y$ define

$$\Omega(E) = \{F \subset X \times Y :\ E \setminus F \in \mathcal{D},\ F \setminus E \in \mathcal{D},\ E \cup F \in \mathcal{D}\}.$$

$\Omega(E)$ also satisfies Lemma 4.1.1(c).

Exercise 4.1.4

Show that $\Omega(E)$ satisfies (c) of Lemma 4.1.1.

Solution [Supplement]**.** If $F_i \in \Omega(E)$ increase to F, then $E \setminus F_i \searrow E \setminus F$, $F_i \setminus E \nearrow F \setminus E$, $E \cup F_i \nearrow E \cup F$, all in the monotone class $\mathcal{D}$, so $F \in \Omega(E)$. Decreasing F_i likewise. □

For $E, F \in \mathcal{E}$, (6) gives $F \in \Omega(E)$, so $\mathcal{E} \subset \Omega(E)$ and $\mathcal{D} \subset \Omega(E)$:

$$E \in \mathcal{E} \implies \mathcal{D} \subset \Omega(E).$$

So for $E \in \mathcal{D}$, $F \in \mathcal{E}$: $E \in \Omega(F)$, i.e. $F \in \Omega(E)$ (symmetry). Hence $\mathcal{E} \subset \Omega(E)$ and $\mathcal{D} \subset \Omega(E)$:

$$E \in \mathcal{D} \implies \mathcal{D} \subset \Omega(E),$$

i.e. $\mathcal{D}$ is closed under differences and finite unions, hence complements ($X \times Y \in \mathcal{D}$), and countable unions $\bigcup_n E_n = \bigcup_n (E_1 \cup \cdots \cup E_n)$. So $\mathcal{D}$ is a σ-algebra, and by (5)

$$\mathcal{D} = \mathcal{C} = \mathcal{S} \times \mathcal{T}. \tag{7}$$

So for $E \in \mathcal{S} \times \mathcal{T} = \mathcal{C}$, the disjoint pieces $E \cap (X_n \times Y_m)$ lie in $\mathcal{B}$, and $E \in \mathcal{B}$ by Lemma 4.1.1(b): $\mathcal{S} \times \mathcal{T} \subset \mathcal{B}$.

Definition (σ-finite)

A measure space $(X, \mathcal{S}, \mu)$ is **σ-finite** if X is a countable union of sets of finite measure.

Theorem 4.1.2

Let $(X,\mathcal{S},\mu)$, $(Y,\mathcal{T},\nu)$ be measure spaces and let $\mathcal{S}\times\mathcal{T}$ be the smallest σ-algebra containing the measurable rectangles. If $E\in\mathcal{S}\times\mathcal{T}$, then $E_x\in\mathcal{T}$ and $E^y\in\mathcal{S}$ for all $x\in X$, $y\in Y$. If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, then the functions φ,ψ of (1) are measurable and (2) holds.

(Proof: the discussion above.) Writing the common value of (2) as $(\mu\times\nu)(E)$,

$$\begin{aligned}(\mu\times\nu)(E) &= \int_X\Big[\int_Y \chi_E(x,y)\,d\nu(y)\Big]d\mu(x) = \int_X \nu(E_x)\,d\mu(x)\\ &= \int_Y\Big[\int_X \chi_E(x,y)\,d\mu(x)\Big]d\nu(y) = \int_Y \mu(E^y)\,d\nu(y).\end{aligned}\tag{8}$$

$\mu\times\nu$ is a measure: for disjoint E_n with union E, $E_x=\bigsqcup_n (E_n)_x$, and by Fact B.25

$$\sum_{n=1}^{\infty}(\mu\times\nu)(E_n)=\sum_{n=1}^{\infty}\int_X \nu((E_n)_x)\,d\mu(x)=\int_X\sum_{n=1}^{\infty}\nu((E_n)_x)\,d\mu(x)=\int_X\nu(E_x)\,d\mu(x)=(\mu\times\nu)(E).$$

$(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is the **product measure** of the σ-finite spaces.

Exercise 4.1.5

Show that $(\mu\times\nu)(S\times T)=\mu(S)\nu(T)$ for every $S\in\mathcal{S}$, $T\in\mathcal{T}$.

Solution [Supplement]. $\nu((S\times T)_x)=\nu(T)\chi_S(x)$, so by (8) $(\mu\times\nu)(S\times T)=\nu(T)\mu(S)$ $(0\cdot\infty=0)$. □

Exercise 4.1.6

If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, show that $(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is σ-finite.

Solution [Supplement]. $X\times Y=\bigsqcup_{m,n}(X_n\times Y_m)$ with $(\mu\times\nu)(X_n\times Y_m)=\mu(X_n)\nu(Y_m)<\infty$ (Exercise 4.1.5). □

4.1.5 The product of Lebesgue measures; Borel sets

For Lebesgue measure $(\mathbb{R},\mathcal{M},m)$: an open $U\subset\mathbb{R}^2$ is the countable union of the

$$\{(a,b)\times(c,d)\subset U:\ a,b,c,d \text{ rational}\}$$

it contains, so open sets lie in $\mathcal{M}\times\mathcal{M}$. With $\mathcal{B}_n$ the smallest σ-algebra containing the open sets of $\mathbb{R}^n$,

$$\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}.$$

Members of $\mathcal{B}_n$ are the **Borel sets** (e.g. closed, G_δ, F_σ sets).

But $\mathcal{M}\times\mathcal{M}$ has a defect: for $\emptyset\neq E\in\mathcal{M}$ with $m(E)=0$ and $F\subset\mathbb{R}$, $F\notin\mathcal{M}$, (3) gives $E\times F\notin\mathcal{M}\times\mathcal{M}$ ($(E\times F)_x=F$ for $x\in E$), though $E\times F\subset E\times\mathbb{R}$, a null set.

Definition (complete measure)

A measure space $(X,\mathcal{S},\mu)$ is **complete** if

$$E\subset F,\ F\in\mathcal{S},\ \mu(F)=0\implies E\in\mathcal{S}.$$

So $(\mathbb{R}^2,\mathcal{M}\times\mathcal{M},m\times m)$ is not complete, while $(\mathbb{R},\mathcal{M},m)$ is (Fact B.2).

Proposition 4.1.3 (completion)

For a measure space $(X, \mathcal{S}, \mu)$ let

$$\mathcal{S}^* = \{E \subset X : \text{ there exist } A, B \in \mathcal{S} \text{ with } A \subset E \subset B,\ \mu(B \setminus A) = 0\},$$

and for $E \in \mathcal{S}^*$ put $\mu^*(E) = \mu(A)$. Then $(X, \mathcal{S}^*, \mu^*)$ is a complete measure space.

Proof. *Well-definedness.* If $A_i \subset E \subset B_i$, $\mu(B_i \setminus A_i) = 0$ $(i = 1, 2)$, then

$$A_2 \setminus A_1 \subset E \setminus A_1 \subset B_1 \setminus A_1,$$

so $\mu(A_2 \setminus A_1) = 0$, likewise $\mu(A_1 \setminus A_2) = 0$, and $\mu(A_1) = \mu(A_2)$.

$\mathcal{S}^*$ is a σ-algebra. $\emptyset \in \mathcal{S}^*$. If $A \subset E \subset B$, then $B^c \subset E^c \subset A^c$ with $A^c \setminus B^c = B \setminus A$. For $A_n \subset E_n \subset B_n$, $\mu(B_n \setminus A_n) = 0$, the unions $A \subset E \subset B$ satisfy

$$B \setminus A = \bigcup_n (B_n \setminus A) \subset \bigcup_n (B_n \setminus A_n),$$

so $\mu(B \setminus A) = 0$.

μ^* is a measure. Disjoint E_n have disjoint A_n, so

$$\mu^*(E) = \mu(A) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \mu^*(E_n).$$

Completeness. If $E \subset F$, $\mu^*(F) = 0$, take $A \subset F \subset B$ with $\mu(B) = 0$; then $\emptyset \subset E \subset B$, so $E \in \mathcal{S}^*$. ■

Exercise 4.1.7

If $(X, \mathcal{S}, \mu)$ is complete, show that $\mathcal{S}^* = \mathcal{S}$.

Solution [Supplement]. $\mathcal{S} \subset \mathcal{S}^*$ $(A = B = E)$. Conversely, if $A \subset E \subset B$, $\mu(B \setminus A) = 0$, then $E \setminus A \subset B \setminus A$ is in $\mathcal{S}$ by completeness, and $E = A \cup (E \setminus A) \in \mathcal{S}$. □

4.1.6 Lebesgue measure on $\mathbb{R}^2$

Compare $\mathcal{B}_2^*$ and $(\mathcal{M} \times \mathcal{M})^*$; clearly $\mathcal{B}_2^* \subset (\mathcal{M} \times \mathcal{M})^*$.

For $S \in \mathcal{M}$, Fact B.3 gives an F_σ-set A and a G_δ-set B with $A \subset S \subset B$, $m(B \setminus A) = 0$; $A \times \mathbb{R}$, $B \times \mathbb{R} \in \mathcal{B}_2$ and

$$(m \times m)[(B \times \mathbb{R}) \setminus (A \times \mathbb{R})] = (m \times m)[(B \setminus A) \times \mathbb{R}] = 0,$$

so $S \times \mathbb{R} \in \mathcal{B}_2^*$; likewise $\mathbb{R} \times T \in \mathcal{B}_2^*$, hence $S \times T \in \mathcal{B}_2^*$. So $\mathcal{M} \times \mathcal{M} \subset \mathcal{B}_2^*$, and $(\mathcal{M} \times \mathcal{M})^* \subset \mathcal{B}_2^{**} = \mathcal{B}_2^*$ gives

$$(\mathcal{M} \times \mathcal{M})^* = \mathcal{B}_2^*.$$

Write $\mathcal{M}_2$, $m_2 = (m \times m)^*$: the complete space $(\mathbb{R}^2, \mathcal{M}_2, m_2)$ is **Lebesgue measure on** $\mathbb{R}^2$. (The $\mathbb{R}^1$ method also works directly on $\mathbb{R}^n$.)

Exercise 4.1.8

Use induction to define Lebesgue measure on $\mathbb{R}^n$.

Solution [Supplement]. Given complete $(\mathbb{R}^{n-1}, \mathcal{M}_{n-1}, m_{n-1})$ $(m_1 = m)$, σ-finite as $\mathbb{R}^{n-1} = \bigcup_k [-k,k]^{n-1}$, identify $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, form $(\mathbb{R}^n, \mathcal{M}_{n-1} \times \mathcal{M}, m_{n-1} \times m)$ (Theorem 4.1.2), and set

$$\mathcal{M}_n = (\mathcal{M}_{n-1} \times \mathcal{M})^*, \qquad m_n = (m_{n-1} \times m)^*$$

(Proposition 4.1.3). As above, $\mathcal{B}_n \subset \mathcal{M}_{n-1} \times \mathcal{M}$ (rational boxes) and $\mathcal{M}_n = \mathcal{B}_n^*$ (each $S \in \mathcal{M}_{n-1}$ lies between an F_σ- and a G_δ-set differing by a null set). By Exercise 4.1.5, $m_n(\prod_i [a_i, b_i]) = \prod_i (b_i - a_i)$. □

4.2 Fubini's Theorem

Motivation. Tonelli swaps the order of integration for nonnegative functions, Fubini for integrable ones; counterexamples show a hypothesis is needed.

(8) swaps the order for characteristic functions; we extend to nonnegative and integrable functions. For f on $X \times Y$ let

$$f_x : y \mapsto f(x,y), \qquad f^y : x \mapsto f(x,y).$$

Proposition 4.2.1

If $f : X \times Y \to \mathbb{R}^*$ is $\mathcal{S} \times \mathcal{T}$-measurable, then for every $x \in X$ and $y \in Y$, f_x is $\mathcal{T}$-measurable and f^y is $\mathcal{S}$-measurable.

Proof. For $\alpha \in \mathbb{R}$, $E = \{f > \alpha\} \in \mathcal{S} \times \mathcal{T}$ has $E_x = \{f_x > \alpha\} \in \mathcal{T}$ by (3); likewise f^y. ■

"Measurable" on $X \times Y$ now means $\mathcal{S} \times \mathcal{T}$-measurable.

Exercise 4.2.1

If $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable with respect to Lebesgue measure on $\mathbb{R}^2$, is $f_x : \mathbb{R} \to \mathbb{R}$ always Lebesgue measurable for every $x \in \mathbb{R}$?

Solution [Supplement]. No. For non-measurable $N \subset \mathbb{R}$, $f = \chi_{\{0\} \times N}$ is $\mathcal{M}_2$-measurable ($\{0\} \times N \subset \{0\} \times \mathbb{R}$, m_2-null, and completeness), but $f_0 = \chi_N$ is not. (Only a.e. x works: Theorem 4.2.6.) □

Theorem 4.2.2 (Tonelli)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite. For every measurable $f : X \times Y \to [0, \infty]$:

(a) the functions $x \mapsto \int_Y f_x \, d\nu$ and $y \mapsto \int_X f^y \, d\mu$ are measurable;

(b)
$$\int_X \Big[\int_Y f_x \, d\nu \Big] d\mu = \int_{X \times Y} f \, d(\mu \times \nu) = \int_Y \Big[\int_X f^y \, d\mu \Big] d\nu.$$

Proof. By Fact B.5 take simple $s_n \nearrow f$ and put

$$\varphi(x) = \int_Y f_x \, d\nu, \qquad \varphi_n(x) = \int_Y (s_n)_x \, d\nu.$$

Characteristic functions satisfy (a), (b) by Theorem 4.1.2, hence so do simple functions. As $(s_n)_x \nearrow f_x$, monotone convergence gives $\varphi_n \nearrow \varphi$, so φ is measurable, and

$$\int_{X \times Y} f \, d(\mu \times \nu) = \lim_{n \to \infty} \int_{X \times Y} s_n \, d(\mu \times \nu) = \lim_{n \to \infty} \int_X \varphi_n \, d\mu = \int_X \varphi \, d\mu.$$

Likewise for the other order. ■

Exercise 4.2.2

Derive Fact B.25 from Tonelli's theorem.

Solution [Supplement]. Fact B.25: $\int_X \sum_n f_n\,d\mu = \sum_n \int_X f_n\,d\mu$ for measurable $f_n \ge 0$. Take $Y = \mathbb{N}$ with counting measure c (σ-finite) and $f(x,n) = f_n(x)$, measurable as $\{f > \alpha\} = \bigcup_n \{f_n > \alpha\} \times \{n\}$. As $\int_{\mathbb{N}} g\,dc = \sum_n g(n)$, Tonelli gives

$$\int_X \sum_n f_n(x)\,d\mu(x) = \int_{X\times\mathbb{N}} f\,d(\mu \times c) = \sum_n \int_X f_n\,d\mu.$$

□

For integrable functions we need a sufficient condition for integrability.

Theorem 4.2.3 (Fubini)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite and let $f : X \times Y \to \mathbb{C}$ be measurable with

$$\int_X \Big[\int_Y |f(x,y)|\,d\nu(y)\Big] d\mu(x) < \infty.$$

Then:

(a) $f \in L^1(\mu \times \nu)$; moreover $f_x \in L^1(\nu)$ for almost every $x \in X$ and $f^y \in L^1(\mu)$ for almost every $y \in Y$;

(b) the functions $x \mapsto \int_Y f_x\,d\nu$ and $y \mapsto \int_X f^y\,d\mu$, defined almost everywhere, are integrable;

(c)
$$\int_X \Big[\int_Y f_x\,d\nu\Big] d\mu = \int_{X\times Y} f\,d(\mu\times\nu) = \int_Y \Big[\int_X f^y\,d\mu\Big] d\nu.$$

Proof. By Tonelli $\int_{X\times Y}|f| < \infty$, so $f \in L^1(\mu\times\nu)$. Take f real ($f = u + iv$) and put

$$\varphi(x) = \int_Y f_x\,d\nu, \qquad \varphi_1(x) = \int_Y (f_+)_x\,d\nu, \qquad \varphi_2(x) = \int_Y (f_-)_x\,d\nu.$$

By Tonelli,

$$\int_X \varphi_1\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) < \infty,$$

so $\varphi_1, \varphi_2 \in L^1(\mu)$, and

$$E = \{x \in X : \varphi_1(x) < \infty,\ \varphi_2(x) < \infty\}$$

has $\mu(E^c) = 0$. For $x \in E$, $f_x = (f_+)_x - (f_-)_x \in L^1(\nu)$ and $\varphi = \varphi_1 - \varphi_2 \in L^1(\mu)$. Tonelli for $f_\pm$ gives

$$\int_X \varphi\,d\mu = \int_X \varphi_1\,d\mu - \int_X \varphi_2\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) - \int_{X\times Y} f_-\,d(\mu\times\nu) = \int_{X\times Y} f\,d(\mu\times\nu).$$

The other half is identical. ■

Exercise 4.2.3

In Fubini's theorem let Y be the set of natural numbers with the measure of Fact B.4 (counting measure). What conclusion do you get? What if X is also taken to be $\mathbb{N}$ in the same way?

Solution [Supplement]. With $Y = \mathbb{N}$, counting measure c, $f_n(x) = f(x,n)$ and $\int_{\mathbb{N}} g\,dc = \sum_n g(n)$: if $\sum_n \int_X |f_n|\,d\mu < \infty$, then $\sum_n f_n(x)$ converges absolutely a.e., is integrable, and

$$\int_X \sum_{n=1}^{\infty} f_n\,d\mu = \sum_{n=1}^{\infty} \int_X f_n\,d\mu.$$

If also $X = \mathbb{N}$, for $a_{mn} = f(m,n)$ with $\sum_m \sum_n |a_{mn}| < \infty$,

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{mn},$$

both absolutely convergent; for $a_{mn} \geq 0$ (Tonelli) no condition is needed. □

Exercise 4.2.4

On $\mathbb{N} \times \mathbb{N}$ consider

$$f(m,n) = \begin{cases} 2 - 2^{-m}, & m = n, \\ -2 + 2^{-m}, & m = n+1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the two iterated sums and see how this relates to the hypotheses of Tonelli's and Fubini's theorems. Do the same for

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}, \qquad x, y \in [1, \infty).$$

Solution [Supplement]. (i) For fixed n, only $m = n, n+1$ contribute: $(2 - 2^{-n}) + (-2 + 2^{-n-1}) = -2^{-n-1}$, so $\sum_n \sum_m f(m,n) = -\frac{1}{2}$. For fixed m: $m = 1$ gives $2 - \frac{1}{2} = \frac{3}{2}$; for $m \geq 2$, $n = m, m-1$ give 0. So $\sum_m \sum_n f(m,n) = \frac{3}{2}$. No contradiction: f changes sign (no Tonelli), and $\sum\sum|f| \geq \sum_n (2 - 2^{-n}) = \infty$ (no Fubini).

(ii) Since $\dfrac{\partial}{\partial y}\dfrac{y}{x^2+y^2} = \dfrac{x^2-y^2}{(x^2+y^2)^2}$,

$$\int_1^{\infty} f(x,y)\,dy = \Big[\frac{y}{x^2+y^2}\Big]_{y=1}^{\infty} = -\frac{1}{1+x^2}, \qquad \int_1^{\infty}\int_1^{\infty} f\,dy\,dx = -\int_1^{\infty}\frac{dx}{1+x^2} = -\frac{\pi}{4}.$$

As $f(y,x) = -f(x,y)$, the other order gives $+\frac{\pi}{4}$; f changes sign and $\iint|f| = \infty$, so neither theorem applies. □

4.2.1 Borel functions

A **Borel function** on $\mathbb{R}^n$ is $\mathcal{B}_n$-measurable; "measurable" means $\mathcal{M}_n$-measurable. Borel functions are measurable. The next result extends Fact B.26(d).

Proposition 4.2.4

If $f : X \to \mathbb{R}$ is measurable with respect to $(X, \mathcal{S}, \mu)$ and $g : \mathbb{R} \to \mathbb{R}$ is a Borel function, then $g \circ f : X \to \mathbb{R}$ is measurable.

Proof. As $g^{-1}(\alpha, \infty) \in \mathcal{B}_1$, show $f^{-1}(E) \in \mathcal{S}$ for $E \in \mathcal{B}_1$. Let

$$\mathcal{A} = \{E \in \mathcal{B}_1 : \ f^{-1}(E) \in \mathcal{S}\}.$$

$\mathcal{A}$ is a σ-algebra (Exercise 4.2.5) containing the open sets ($f^{-1}(a,b) = \{f > a\} \cap \{f < b\}$), so $\mathcal{A} = \mathcal{B}_1$. ■

Exercise 4.2.5

In the proof of Proposition 4.2.4, show that $\mathcal{A}$ is a σ-algebra.

Solution [Supplement]. $f^{-1}(\emptyset)=\emptyset$, $f^{-1}(E^c)=X\setminus f^{-1}(E)$, $f^{-1}(\bigcup_n E_n)=\bigcup_n f^{-1}(E_n)$, and $\mathcal{B}_1$ is a σ-algebra. □

Tonelli–Fubini need $\mathcal{M}\times\mathcal{M}$-measurability, tedious to check directly; Borel functions on $\mathbb{R}^2$ have it ($\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}$), so the next result extends the theorems to all measurable functions.

Proposition 4.2.5

For every measurable function f on $\mathbb{R}^n$ there is a Borel function g with $f=g$ almost everywhere.

Proof. Take $f\ge 0$ and, by Fact B.6, $f=\sum_k c_k\chi_{E_k}$ with $c_k\ge 0$, $E_k\in\mathcal{M}_n$. Choose Borel $A_k\subset E_k\subset B_k$ with $m_n(B_k\setminus A_k)=0$ and put

$$g(x)=\sum_{k=1}^{\infty} c_k\chi_{A_k}(x),\qquad x\in\mathbb{R}^n.$$

g is Borel (Fact B.26), and

$$\{x: f(x)\neq g(x)\}\subset\bigcup_{k=1}^{\infty}(E_k\setminus A_k)\subset\bigcup_{k=1}^{\infty}(B_k\setminus A_k),$$

a null set. ■

Theorem 4.2.6 (Tonelli–Fubini on $\mathbb{R}^2$)

Let f be a measurable function on $\mathbb{R}^2$.

(a) For almost every x and almost every y, f_x and f^y are measurable.

(b) If $0\le f\le\infty$, then

$$\int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dx\,dy=\int_{\mathbb{R}\times\mathbb{R}} f\,dm_2=\int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dy\,dx. \tag{9}$$

(c) If $f:\mathbb{R}^2\to\mathbb{C}$ and $\int_{\mathbb{R}}\int_{\mathbb{R}}|f(x,y)|\,dx\,dy<\infty$, then $f\in L^1(m_2)$; moreover $f_x, f^y\in L^1(m_1)$ for almost every x,y, and (9) holds.

Proof. By Proposition 4.2.5, $f=g+h$ with g Borel and $h=0$ a.e. g is $\mathcal{M}\times\mathcal{M}$-measurable, so it satisfies the theorem by Proposition 4.2.1 and Theorems 4.2.2, 4.2.3 ($\int g\,dm_2=\int g\,d(m\times m)$). It remains to treat h. Let

$$E=\{(x,y)\in\mathbb{R}^2: h(x,y)\neq 0\}.$$

Then $m_2(E)=0$, so $E\subset F\in\mathcal{M}\times\mathcal{M}$ with $(m\times m)(F)=0$, and by (8) $\int_{\mathbb{R}} m(F_x)\,dx=0$, so

$$A=\{x\in\mathbb{R}: m(F_x)>0\}$$

has $m(A)=0$. For $x\notin A$, $m(E_x)\le m(F_x)=0$ (completeness), and $h_x=0$ off E_x, so h_x is measurable and 0 a.e. Likewise h^y, proving (a);

(b) and (c) for h follow as all integrals of h, h_x, h^y (a.e. x,y) vanish. ■

Exercise 4.2.6

Complete the proof of Theorem 4.2.6.

Solution [Supplement]. Write $f=g+h$. (b): for $f\ge 0$ take $g\ge 0$ (replace g by g_+, h by $f-g_+$). For $x\notin A$, $f_x=g_x$ a.e., so $\int\int f\,dy\,dx=\int\int g\,dy\,dx$; and $\int f\,dm_2=\int g\,dm_2=\int g\,d(m\times m)$. Tonelli

for g and the symmetric argument with h^y give (9). (c): by (b) $\int|f|\,dm_2 = \int\int|f|\,dx\,dy < \infty$, so $f \in L^1(m_2)$, $g \in L^1(m \times m)$ and Fubini applies to g; as $f_x = g_x$ a.e. for a.e. x (likewise f^y), integrability of sections and (9) transfer to f. □

Example 1 (the Dirichlet integral)

Fubini for $f(x,t) = e^{-xt}\sin x$ on $[0,A]\times[0,\infty)$: as

$$\frac{1}{x} = \int_0^\infty e^{-xt}\,dt, \qquad x > 0,$$

we have

$$\int_0^1\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_0^1 x\int_0^\infty e^{-xt}\,dt\,dx \le \int_0^1 dx = 1,$$
$$\int_1^A\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_1^A\int_0^\infty e^{-xt}\,dt\,dx = \int_1^A \frac{1}{x}\,dx < \infty,$$

(with $|\sin x| \le x$ on $[0,1]$), Fubini applies:

$$\int_0^A \frac{\sin x}{x}\,dx = \int_0^A\int_0^\infty e^{-xt}\sin x\,dt\,dx = \int_0^\infty\int_0^A e^{-xt}\sin x\,dx\,dt$$
$$= \int_0^\infty \Big[\frac{e^{-At}}{1+t^2}(-t\sin A - \cos A) + \frac{1}{1+t^2}\Big]dt.$$

As

$$\int_0^\infty \Big|\frac{e^{-tA}}{1+t^2}(-t\sin A - \cos A)\Big|dt \le \int_0^\infty 2e^{-tA}\,dt = \frac{2}{A}$$

($|t\sin A + \cos A| \le 1 + t$, $\frac{1+t}{1+t^2} \le 2$),

$$\lim_{A\to\infty}\int_0^A \frac{\sin x}{x}\,dx = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

4.3 Convolution of Integrable Functions

Motivation. Convolution averages f with weights g; it smooths, becomes multiplication under the Fourier transform, and approximates L^p functions; Fubini makes it well defined.

$L^1(X,\mu)$ carries a product when X has an operation and μ is invariant (Fact B.1); we treat only $\mathbb{R}$.

4.3.1 Definition

For measurable $f, g : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$ consider

$$\int_{\mathbb{R}} f(y)g(x-y)\,dy.$$

By Proposition 4.2.5 take f, g Borel. $(x,y) \mapsto y$ and $(x,y) \mapsto x-y$ are continuous, so by Proposition 4.2.4 and Fact B.26(a)

$$(x,y) \mapsto f(y)g(x-y) \tag{10}$$

is Borel (for measurable f, g, still measurable). By translation invariance (Fact B.1),

$$\int_{\mathbb{R}}\Big[\int_{\mathbb{R}}|f(y)g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\Big[\int_{\mathbb{R}}|g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\,dy\int_{\mathbb{R}}|g(x)|\,dx = \|f\|_1\|g\|_1,$$

so by Fubini

$$x \mapsto \int_{\mathbb{R}} f(y)g(x-y)\,dy \tag{11}$$

is defined a.e., with

$$\int_{\mathbb{R}} \Big| \int_{\mathbb{R}} f(y)g(x-y)\,dy \Big| dx \le \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} |f(y)g(x-y)|\,dx \Big] dy = \|f\|_1 \|g\|_1.$$

Definition (convolution)

For $f, g \in L^1(\mathbb{R})$, (11) is the **convolution** $f * g \in L^1(\mathbb{R})$, and

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{R}). \tag{12}$$

Proposition 4.3.1

For $f, g, h \in L^1(\mathbb{R})$ and constants $\alpha, \beta \in \mathbb{C}$:

(a) $[(f * g) * h](x) = [f * (g * h)](x)$ for almost every x;

(b) $(f * g)(x) = (g * f)(x)$ for every x (for which either side is defined);

(c) $[(\alpha f + \beta g) * h](x) = [\alpha(f * h) + \beta(g * h)](x)$ for almost every x.

Proof. (a) Fix x. Then

$$[(f * g) * h](x) = \int_{\mathbb{R}} (f * g)(y)h(x-y)\,dy = \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} f(z)g(y-z)\,dz \Big] h(x-y)\,dy,$$

and

$$\begin{aligned}[f * (g * h)](x) &= \int_{\mathbb{R}} f(z)(g * h)(x-z)\,dz = \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y)h(x-z-y)\,dy \Big] dz \\ &= \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y-z)h(x-y)\,dy \Big] dz\end{aligned}$$

$(y \mapsto y - z)$. Now

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |f(z)g(y-z)h(x-y)|\,dy\,dz = [(|f| * |g|) * |h|](x),$$

is finite for a.e. x ($|f| * |g|$, $|h| \in L^1$); for such x apply Fubini to $(y, z) \mapsto f(z)g(y-z)h(x-y)$.

(b) Directly,

$$(f * g)(x) = \int_{\mathbb{R}} f(y)g(x-y)\,dy = \int_{\mathbb{R}} f(-y)g(x+y)\,dy = \int_{\mathbb{R}} f(x-y)g(y)\,dy = (g * f)(x),$$

by reflection and translation invariance.

(c) For a.e. x, $y \mapsto f(y)h(x-y)$ and $g(y)h(x-y)$ are in L^1; use linearity (Fact B.10). ■

Definition (Banach algebra)

A Banach space X with a multiplication $(x, y) \mapsto xy$ that is associative, distributive and satisfies

$$\|xy\| \le \|x\|\|y\|, \qquad x, y \in X,$$

is a **Banach algebra**. By (12) and Proposition 4.3.1(a),(c), $L^1(\mathbb{R})$ is one under convolution.

Proposition 4.3.2

If g is differentiable and $f, g, g' \in L^1(\mathbb{R})$, then $f * g$ is differentiable and

$$(f * g)' = f * g'.$$

Exercise 4.3.1

Prove Proposition 4.3.2.

Solution [Supplement]. *Step 1: g is bounded and $g(b) - g(a) = \int_a^b g'$.* An everywhere differentiable function with integrable derivative satisfies the fundamental theorem of calculus (Ch. 5): $g(x) = g(0) + \int_0^x g'$. As $g' \in L^1$, $g(\pm\infty)$ exist, and they are 0 as $g \in L^1$; so g is bounded and $(f * g)(x)$ is defined for *every* x.

Step 2: an integral formula. For $x \in \mathbb{R}$, $h \neq 0$,

$$(f * g)(x + h) - (f * g)(x) = \int_{\mathbb{R}} f(y) \int_0^h g'(x - y + t)\, dt\, dy = \int_0^h (f * g')(x + t)\, dt,$$

by Fubini, which applies because $\int_{\mathbb{R}} \int_0^h |f(y)g'(x - y + t)|\, dt\, dy \leq |h|\, \|f\|_1 \|g'\|_1 < \infty$.

Step 3. So $\frac{(f*g)(x+h)-(f*g)(x)}{h}$ is the average of $f * g'$ between x and $x + h$, which tends to $(f * g')(x)$ wherever $f * g'$ is continuous (everywhere if g' is bounded, by Exercise 4.3.2). In general Step 2 makes $f * g$ absolutely continuous with $(f * g)' = f * g'$ a.e. (Lebesgue differentiation, Ch. 5).

Alternative under bounded g'. If $|g'| \leq M$, the difference quotient $f(y)\frac{g(x+h-y)-g(x-y)}{h}$ is bounded by $M|f(y)|$ and tends to $f(y)g'(x - y)$; dominated convergence gives $(f * g)'(x) = (f * g')(x)$ for every x. □

4.3.2 Convolution of L^p and L^q functions

Let $p, q > 1$ be conjugate exponents. For a function g and $s \in \mathbb{R}$ write $g_s(t) = g(t - s)$. If $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$, Hölder's inequality gives

$$\int_{\mathbb{R}} |f(y)g(x - y)|\, dy = \int_{\mathbb{R}} |f(y)g_{-x}(-y)|\, dy \leq \|f\|_p \|g_{-x}\|_q = \|f\|_p \|g\|_q.$$

Hence $(f * g)(x)$ is defined for every $x \in \mathbb{R}$ and $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$.

Exercise 4.3.2

If $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, show that $f * g$ is uniformly continuous.

Solution [Supplement]. Using $f * g = g * f$, for $x, y \in \mathbb{R}$,

$$\begin{aligned} |(f * g)(x) - (f * g)(y)| &= \left| \int_{\mathbb{R}} g(z)(f(x - z) - f(y - z))dz \right| \\ &\leq \|g\|_\infty \int_{\mathbb{R}} |f(x - z) - f(y - z)|\, dz = \|g\|_\infty \|f_{x-y} - f\|_1, \end{aligned}$$

after the substitution $w = y - z$. By the continuity of translation in L^1 (Fact B.23), $\|f_s - f\|_1 \to 0$ as $s \to 0$. Hence for $\epsilon > 0$ there is $\delta > 0$ with $|(f * g)(x) - (f * g)(y)| < \epsilon$ whenever $|x - y| < \delta$. □

Proposition 4.3.3

Let $p, q \in (1, \infty)$ be conjugate exponents. Then $f * g \in C_0(\mathbb{R})$ for every $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$.

Proof. By Hölder's inequality again, for $x, y \in \mathbb{R}$

$$|(f * g)(x) - (f * g)(y)| \le \int_{\mathbb{R}} |f(z)(g_{-x}(-z) - g_{-y}(-z))|\, dz \le \|f\|_p \|g_{-x} - g_{-y}\|_q,$$

so by Fact B.23 (continuity of translation in L^q, $q < \infty$), $f * g$ is continuous. Next, by Fact B.22 there are sequences $\langle f_n \rangle, \langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\lim_{n\to\infty} \|f_n - f\|_p = 0, \qquad \lim_{n\to\infty} \|g_n - g\|_q = 0.$$

If the supports of f_n and g_n lie in $[-A, A]$ and $[-B, B]$, then the support of $f_n * g_n$ lies in $[-A - B, A + B]$, so $\langle f_n * g_n \rangle$ is a sequence in $C_c(\mathbb{R})$. Then

$$\begin{aligned} \|f_n * g_n - f * g\|_\infty &\le \|f_n * g_n - f_n * g\|_\infty + \|f_n * g - f * g\|_\infty \\ &\le \|f_n\|_p \|g_n - g\|_q + \|f_n - f\|_p \|g\|_q \to 0, \end{aligned}$$

so $\langle f_n * g_n \rangle$ converges uniformly to $f * g$. A uniform limit of functions in $C_c(\mathbb{R})$ belongs to $C_0(\mathbb{R})$. ■

4.3.3 Averaging kernels

It is usually hard to guess what $f * g$ looks like, but sometimes one can. Consider

$$k_r = \frac{1}{2r} \chi_{[-r,r]}, \qquad r > 0. \tag{13}$$

By definition

$$(k_r * f)(x) = \int_{-\infty}^{\infty} k_r(x - y) f(y)\, dy = \frac{1}{2r} \int_{x-r}^{x+r} f(y)\, dy,$$

the average of f over the neighborhood $[x - r, x + r]$ of x. So $k_r * f$ should be better behaved than f and approach f as $r \to 0$.

$k_r * f$ is uniformly continuous for $f \in L^1(\mathbb{R})$. By Fact B.15 (absolute continuity of the integral), for any $\epsilon > 0$ there is $\delta \in (0, 2r)$ with

$$E \in \mathcal{M},\ m(E) \le \delta \implies \int_E |f(t)|\, dt < 2r\epsilon.$$

If $|x - y| < \frac{\delta}{2} < r$, let $E = [x - r, x + r] \triangle [y - r, y + r]$, where $A \triangle B = (A \cup B) \setminus (A \cap B)$; then $m(E) = 2|x - y| \le \delta$. Therefore

$$|(k_r * f)(x) - (k_r * f)(y)| = \frac{1}{2r} \Big| \int_{x-r}^{x+r} f(t)\, dt - \int_{y-r}^{y+r} f(t)\, dt \Big| \le \frac{1}{2r} \int_E |f(t)|\, dt < \epsilon,$$

so $f * k_r$ is uniformly continuous on $\mathbb{R}$.

Uniform approximation. If f is uniformly continuous, for each $\epsilon > 0$ there is $\delta > 0$ with $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. If $r < \delta$, then

$$|f(x) - (f * k_r)(x)| \le \frac{1}{2r} \int_{x-r}^{x+r} |f(x) - f(t)|\, dt < \epsilon,$$

so $f * k_r \to f$ uniformly as $r \to 0$.

Exercise 4.3.3

For $f = \chi_{[-1,1]}$ sketch $x \mapsto (f * k_r)(x)$ for $r = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$. Do the same for $f(x) = \max\{1 - |x|, 0\}$.

Solution [Supplement]. *(i)* $f = \chi_{[-1,1]}$. $(f * k_r)(x) = \frac{1}{2r}\, m([x-r, x+r] \cap [-1,1])$, an even function. For $r \le 1$:

$$(f * k_r)(x) = \begin{cases} 1, & |x| \le 1 - r, \\ \dfrac{1 + r - |x|}{2r}, & 1 - r \le |x| \le 1 + r, \\ 0, & |x| \ge 1 + r. \end{cases}$$

A trapezoid: height 1 on $[-(1-r), 1-r]$, linear ramps of width $2r$ through $\frac{1}{2}$ at ± 1. For $r = 1$ it is the triangle $(2 - |x|)/2$ on $[-2, 2]$; as $r \to 0$ it tends to $\chi_{[-1,1]}$ pointwise (to $\frac{1}{2}$ at ± 1), not uniformly, f being discontinuous.

(ii) $f(x) = \max\{1 - |x|, 0\}$ *(tent).* For $r \le \frac{1}{2}$ and $x \ge 0$ (the function is even), a direct integration gives

$$(f * k_r)(x) = \begin{cases} 1 - \dfrac{x^2 + r^2}{2r}, & 0 \le x \le r, \\ 1 - x, & r \le x \le 1 - r, \\ \dfrac{(1 + r - x)^2}{4r}, & 1 - r \le x \le 1 + r, \\ 0, & x \ge 1 + r. \end{cases}$$

The tent's corners at $0, \pm 1$ become parabolic arcs of width $2r$: the result is C^1 with maximum $1 - \frac{r}{2}$, and $\|f * k_r - f\|_\infty = \frac{r}{2} \to 0$. For $r = 1$: $(f * k_1)(x) = \frac{1}{2} - \frac{x^2}{4}$ for $|x| \le 1$ and $\frac{(2-|x|)^2}{4}$ for $1 \le |x| \le 2$. □

Exercise 4.3.4

Let $f_0 = \chi_{[-1,1]}$ and $f_{n+1} = f_n * f_0$ $(n = 0, 1, 2, \dots)$. Sketch how the graphs of this sequence change.

Solution [Supplement]. $f_1(x) = m([x-1, x+1] \cap [-1,1]) = \max\{2 - |x|, 0\}$, a triangle of height 2 on $[-2, 2]$. Inductively, since $(f_n * f_0)(x) = \int_{x-1}^{x+1} f_n$:

- f_n is even, nonnegative, supported in $[-(n+1), n+1]$, increasing on $(-(n+1), 0]$;
- f_n is a piecewise polynomial of degree n with breakpoints at the integers of the same parity as $n + 1$, and $f_n \in C^{n-1}$ (each convolution with f_0 gains one derivative: $f'_{n+1}(x) = f_n(x + 1) - f_n(x - 1)$);
- $\int f_n = \left(\int f_0\right)^{n+1} = 2^{n+1}$ by Fubini.

So the graphs go step → triangle → quadratic bell → $\cdots$, smoother and wider: $f_n / 2^{n+1}$ is the density of a sum of $n + 1$ independent uniforms on $[-1, 1]$ (variance $(n+1)/3$), so by the CLT it approaches $\frac{2^{n+1}}{\sqrt{2\pi(n+1)/3}} e^{-3x^2/(2(n+1))}$. □